



Savitribai Phule Pune University

T. Y. B.Sc Cyber and Digital Science

Semester-VI

CDS-364 Lab on CDS-361

Digital Forensics-2

Lab Book

Name: _____

College Name: _____

Roll No.: _____ **Division:** _____

Academic Year : _____

CERTIFICATE

This is to certify that Mr./Ms. _____

Seat Number _____ of **T.Y.BSC (Cyber and Digital Science) Sem-VI** has successfully completed Laboratory course
(Digital Forensic 2) in the year ____.

He/ She has scored _____ mark out of 10 (For Lab Book).

Subject Teacher

H.O.D./Coordinator

Internal Examiner

External Examiner

Introduction

1. About the Workbook:

This workbook is intended to be used by T.Y.B.Sc (Cyber and Digital Science) Semester VI students for (Digital Forensic 2) Assignments. This workbook is designed by considering all the practical concepts / topics mentioned in syllabus.

2. The objectives of this Workbook are:

- 1) Defining the scope of the course.
- 2) To bring the uniformity in the practical conduction and implementation in all colleges affiliated to SPPU.
- 3) To have continuous assessment of the course and students.
- 4) Providing ready reference for the students during practical implementation.
- 5) Provide more options to students so that they can have good practice before facing the examination.
- 6) Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. How to use this Workbook:

The workbook contains Eight assignments

It is mandatory for students to complete all the assignments in given slot.

4. Instructions to the students:

Please read the following instructions carefully and follow them.

- 1) Students are expected to carry this workbook every time they come to the lab for practical.
 - 2) Students should prepare for the assignment by reading the relevant material which is mentioned in ready reference.
 - 3) Instructor will specify which problems to solve in the lab during the allotted slot and student should complete them and get verified by the instructor. However, student should spend additional hours in Lab and at home to cover all workbook assignments if needed.
- Students will be assessed for each assignment on a scale from 0 to 5

Not done	0
Incomplete	1
Late Complete	2
Needs improvement	3
Complete	4
Well Done	5

5. Instruction to the Instructors:

Make sure that students should follow above instructions.

- Explain the assignment and related concepts using white board if required or by demonstrating the software.
- Give specific input to fill the blanks in queries which can vary from student to student.
- Evaluate each assignment carried out by a student on a scale of 5 as specified above by ticking appropriate box.
- The value should also be entered on assignment completion page of the respective Lab course.

6. Instructions to the Lab administrator:

You have to ensure appropriate hardware and software is made available to each student.

Editors:

- 1) Mr. Satyavan Kunjir**
- 2) Mr. Sammed Bukshete**
- 3) Mrs. Renuka Kulkarni**

Dr. D. Y. Patil ACS College, Pimpri, Pune
Dr. D. Y. Patil ACS College, Pimpri, Pune
Dr. D. Y. Patil ACS College, Pimpri, Pune

Reviewed By:

- 1) Dr. Ranjit D. Patil**
- 2) Dr. Sujata Patil**

Dr. D. Y. Patil ACS College, Pimpri, Pune
Dr. D. Y. Patil ACS College, Pimpri, Pune

Assignment Index

Sr. No.	Assignment Name	Page Number
1	Inspecting Emails	1
2	Accessing mobile data	4
3	Extracting social media data	6
4	Accessing mobile device data	8
5	Recovering deleted files on mobile device	10
6	Multimedia forensics	12
7	Cloud Forensics	15
8	Virtual Machine and Network Forensics	17

Signature of Instructor:

Assignment Completion Sheet

Sr. No.	Assignment Name	Marks (out of 5)	Teacher's Sign
1	Inspecting Emails		
2	Accessing mobile data		
3	Extracting social media data		
4	Accessing mobile device data		
5	Recovering deleted files on mobile device		
6	Multimedia forensics		
7	Cloud Forensics		
8	Virtual Machine and Network Forensics		
Total (Out of 40)			
Total (Out of 5)			

Signature of Instructor:

Assignment 1

Inspecting Emails

Objective:

Understand the structure of email headers and learn to analyze various components like servers, domain keys, and attachments.

Theory:

NEED FOR E-MAIL FORENSIC INVESTIGATION

Cybercriminals spoof e-mail messages to carry out various illegitimate activities through e-mail system and remain underground to evade any possible legal action against them. These include

- abuses like spamming, phishing, cyber bullying, child pornography, sexual harassment, racial vilification, etc.,
- misuse by transmitting viruses, worms, Trojan horses, hoaxes, and other malicious programs with an intent to spread them over Internet, and
- carry out Internet infrastructure crimes through Denial of Services and Directory Harvesting Attacks.

Senders can lie about their true identities in various ways by using or misusing different techniques that include:

- **Spoofing:** It is an attempt to conceal the source of an e-mail message by placing false information in its headers. All possibilities to lie about true identities of sender listed above lead to spoofing. E-mail spoofing may be combined with IP spoofing to make its detection very difficult.
- **Unauthorized Networks:** Wired or wireless networks that have been compromised by perpetrators to gain unauthorized access to the Internet can disguise their identities while sending an e-mail.
- **Open Mail Relays:** An open mail relay is a misconfigured mail relay that accepts mail from any computer and forwards it to another computer which otherwise should have accepted mail for and from specific computers. Such a relay becomes vulnerable to spammers and phishers who hide their identities behind these relays.
- **Annominizers or re-mailers:** Re-mailers are websites that operate under the guise of protecting privacy of Internet users offering anonymous Internet surfing. They intentionally strip headers from e-mail and some even do not maintain server logs.
- **Open Proxy:** A proxy server is a machine that allows computers to connect through it to some other computer on the Internet. HTTP proxy server provided by ISPs, Corporate Proxy Server, transparent proxy server, and Open proxy server also called anonymous proxy server are different types of proxy servers which provide different levels of anonymity. Users connecting to Internet through these proxy servers share IP address. An Open proxy server does not maintain a strict log of user activities unlike others which maintain user logs synchronized with reliable time servers. Such open proxy servers provide anonymity and untraceable Internet activity.

- **SSH Tunnel or Port-Redirector:** A Tunnel in Internet means a secure data path through an un-trusted network. Depending upon the software and techniques used, tunneling can be accomplished through many ways

ANALYZING AN E-MAIL MESSAGE

Table 1: Sample header set of an e-mail message

<i>Para No</i>	<i>Header</i>	<i>Value</i>
1	X-Apparently-To:	<i>bob@bob.com via a4.b4.c4.d4; Tue, 30 Nov 2010 07:36:34 -0800</i>
2	Return-Path:	<i>< alice@alice.com ></i>
3	Received-SPF:	<i>none (mta1294.mail.mud.bob.com: domain of alice@alice.com does not designate permitted sender hosts)</i>
4	X-Spam-Ratio:	<i>3.2</i>
5	X-Originating-IP:	<i>[a2.b2.c2.d2]</i>
6	X-Sieve:	<i>CMU Sieve 2.3</i>
7	X-Spam-Charsets:	<i>Plain='utf-8' html='utf-8'</i>
8	X-Resolved-To:	<i>bob@bob.com</i>
9	X-Delivered-To:	<i>bob@bob.com</i>
10	X-Mail-From:	<i>alice@alice.com</i>
11	Authentication-Results:	<i>mta1294.mail.mud.bob.com from=alice.com; domainkeys=neutral (no sig); from=alice.com; dkim=neutral (no sig)</i>
12	Received:	<i>from 127.0.0.1 (EHLO mailbox-us-s-7b.tariq.com) (a2.b2.c2.d2) by mta1294.mail.mud.bob.com with</i>

Exercise:

1. Open an email in a web-based email service (e.g., Gmail or Yahoo).
2. Click on the down arrow next to the Reply arrow and choose "Show original."
3. Download the original and inspect the header to identify different components.

Questions:

1. What is the name and file type of the base64 encoded attachment?
2. To which email was this file attached? Identify two pieces of data that tell you that.
3. Who was this email sent from, to, and when was it sent?

4. In what time zone does the computer that was used to send the email reside (presuming it corresponds to the time zone setting for the computer)? What data in the header tells you that?
5. In what time zone does the sender's mail server reside (presuming it corresponds to the time zone setting for the computer)? What data in the header tells you that?
6. If you believed the source email address information has been spoofed, which specific IP address would you resolve, and subsequently contact the owner of, to find out who might have really sent the email?

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 2

Accessing Mobile Data

Objective:

Utilize ADB and Autopsy to extract and analyze information from an Android device for digital forensics.

Theory:

Mobile Device Forensics

Mobile device forensics, also known as mobile forensics, is a subfield of digital forensics that involves extracting information from a mobile device (such as smartphones and tablets) in a forensically sound manner. The information obtained via mobile device forensics may include deleted files, application data, GPS data, call logs, text messages, and photographs and videos.

Like other domains of forensics, mobile device forensics is commonly used to recover evidence in connection with a criminal investigation. As such, mobile device forensic investigators must take care to retrieve and analyze data that is legally admissible as evidence.

Mobile device forensics has connections with other branches of digital forensics—such as network forensics, computer forensics, and malware analysis—in terms of the knowledge and skill set required. However, the distinguishing feature of mobile device forensics is that the extracted data is located on a mobile device.

Therefore, mobile device forensic analysts must be intimately familiar with mobile devices and their operating systems and file systems. They should also have experience with various software and hardware tools for extracting data from mobile devices. Finally, mobile device forensic analysts should have strong problem-solving and critical thinking skills and knowledge of the legal issues surrounding collecting data from mobile devices.

The Process of Mobile Device Forensics

There are four general steps to follow during a forensic investigation: identifying the evidence, acquiring the evidence, analyzing the evidence, and producing a forensic report. Below are these four steps as they pertain to the process of mobile device forensics:

1. **Device seizure:** First, the mobile device is seized from its user. At this stage, investigators should also start documenting the chain of custody. For example, the records of who handled the device and when. A search warrant is usually required if the device is used in a criminal investigation.
2. **Device acquisition:** Investigators create a sector-level duplicate of the device, a process known as “imaging” or “acquisition.” This duplicate image and the original device are passed through a hashing function, and their outputs are compared to ensure that it is an exact copy. Next, analysts decide on the investigation’s proper approach and goals.
3. **Device analysis:** Investigators begin work on the device image to confirm a hypothesis or search for hidden data. Specialized tools (such as those described in the next section) are used to help find and recover information. Data may be located

within the accessible hard disk space, deleted (unallocated) disk space, or the operating system cache.

4. **Reporting:** After acquiring the data, investigators store and analyze it to reconstruct a plausible version of events. A report is prepared, which may be technical or non-technical, depending on the audience.

Mobile Device Forensics: Tools and Techniques

Mobile device forensic analysts use various tools and techniques to analyze devices. For example, there are multiple ways to extract information from a mobile device:

- **Logical extraction:** The device is connected to a forensics workstation via a hardware cable or a protocol such as Bluetooth. This approach is quick and relatively straightforward but also the most limited. Logical extraction tools include Oxygen Forensic Device Extractor and XRY Logical.
- **Physical extraction (hex dump):** The device's flash memory is copied bit by bit. This approach is the most extensive but technically complex and dependent on the manufacturer. Physical extraction tools include Cellebrite UFED Physical Pro and XRY Physical.

Exercise:

1. Extracting device information of your Android device using ADB.
2. Use the SQLite Browser to display call logs.
3. Display all call logs using ADB.
4. View browser history information using Autopsy.

Questions:

1. What information can be extracted using ADB for device identification?
2. How can call logs be displayed using the SQLite Browser?
3. What ADB command can be used to view all call logs from an Android device?
4. What are the key steps to view browser history information using Autopsy?

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 3

Extracting Social Media Data

Objective:

Employ techniques to extract and analyze data from social media applications, focusing on platforms like Facebook or Whats App.

Theory:

Extracting data from social media applications involves understanding the structure of their databases and employing appropriate tools to access and analyze the information stored.

Tools that collect social media data:

NCapture plug-in for NVivo

- NCapture is a free web-browser extension for Chrome that enables researchers to gather web content--including data from YouTube, LinkedIn, websites, and online PDFs--to import into NVivo.
- When capturing data from Facebook for the first time, researchers need to authorize NCapture.
- Email datahelp@gmu.edu for help with NVivo.
- Due to the changes made to Twitter's API, NCapture can no longer function as designed with Twitter data.

Advanced collection methods:

- **Facebook Graph API**

- The Graph API is the primary way to get data into and out of the Facebook platform. It is an HTTP-based API that apps can use to programmatically query data, post new stories, manage ads, upload photos, and perform a wide variety of other tasks.
- Researchers need to register as a Facebook Developer and create a Facebook App.

- **Instagram Basic Display API**

- The Instagram Basic Display API allows users of your app to get basic profile information, photos, and videos in their Instagram accounts. The API can be used to access any type of Instagram account but only provides read-access to basic data.
- View their documentation on getting started.

Other resources:

- **Data Knowledge Hub for Monitoring Online Discourse**

- The Data Knowledge Hub for Monitoring Online Discourse (Data Knowledge Hub) is an initiative that aims to provide a central resource for researchers, social scientists, data scientists, journalists, practitioners, and policy makers interested in monitoring social media and online discourse more broadly.

- **Social Media Research Toolkit**

- The Social Media Research Toolkit is a list of 50+ social media research tools curated by researchers at the Social Media Lab at Ted Rogers School of Management, Ryerson University. The kit features tools that have been used in peer-reviewed academic studies. Many tools are free to use and require little or no programming.

Exercise:

Extract data from any social media application (e.g., Facebook, WhatsApp) and display relevant information.

Questions:

1. What steps did you follow to extract data from the chosen social media application?
2. What types of relevant information were you able to retrieve?
3. Are there any challenges or limitations in extracting data from social media platforms?

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 4

Accessing Mobile Device Data

Objective:

Learn to create an image of mobile device data and inspect the contents, making detailed observations.

Theory:

Creating an image of mobile device data involves capturing a snapshot of the device's storage, preserving its current state. Inspecting the contents allows for detailed analysis and observation of the stored data.

Steps in the mobile forensics process

- Digital forensics operates on the principle that evidence should always be adequately preserved, processed, and admissible in a court of law. Some legal considerations go hand in hand with the confiscation of mobile devices.
- There are two major risks concerning this phase of the mobile forensic process: Lock activation (by user/suspect/inadvertent third party) and Network / Cellular connection.
- Network isolation is always advisable, and it could be achieved either through 1) Airplane Mode + Disabling Wi-Fi and Hotspots, or 2) Cloning the device SIM card.

Airplane mode

Mobile devices are often seized switched on; and since the purpose of their confiscation is to preserve evidence, the best way to transport them is to attempt to keep them turned on to avoid a shutdown, which would inevitably alter files.

Phone jammer Faraday bag

A Faraday box/bag and external power supply are common types of equipment for conducting mobile forensics. While the former is a container specifically designed to isolate mobile devices from network communications and, at the same time, help with the safe transportation of evidence to the laboratory, the latter, is a power source embedded inside the Faraday box/bag. Before putting the phone in the Faraday bag, disconnect it from the network, disable all network connections (Wi-Fi, GPS, Hotspots, etc.), and activate the flight mode to protect the integrity of the evidence.

Exercise:

1. Create an image of your phone data using appropriate tools (e.g., ADB or other forensic tools).
2. Inspect the contents of the created image and note your observations.

]Questions:

1. Which tool or method did you use to create an image of your phone data, and why?
2. What observations can you make regarding the structure and organization of the data in the image?
3. Are there any specific challenges or interesting findings during the inspection process?

Assignment Evaluation**0: Not Done []****1: Incomplete []****2: Late Complete []****3: Need Improvement []****4: Complete []****5: Well Done []****Signature of Instructor**

Assignment 5

Recovering Deleted Files on Mobile Device

Objective:

Develop skills in recovering deleted files from an Android device using tools like Autopsy and utilizing platform-specific methods for SMS and contacts recovery.

Theory:

Recovering deleted files involves using forensic tools and platform-specific methods to analyze and restore data that has been deleted from a mobile device.

The data recovery process varies, depending on the circumstances of the data loss, the data recovery software used to create the backup and the backup target media. For example, many desktop and laptop backup software platforms allow users to restore lost files themselves, while restoration of a corrupted database from a tape backup is a more complicated process that requires IT intervention. Data recovery services can also be used to retrieve files that were not backed up and accidentally deleted from a computer's file system, but still remain on the hard disk in fragments.

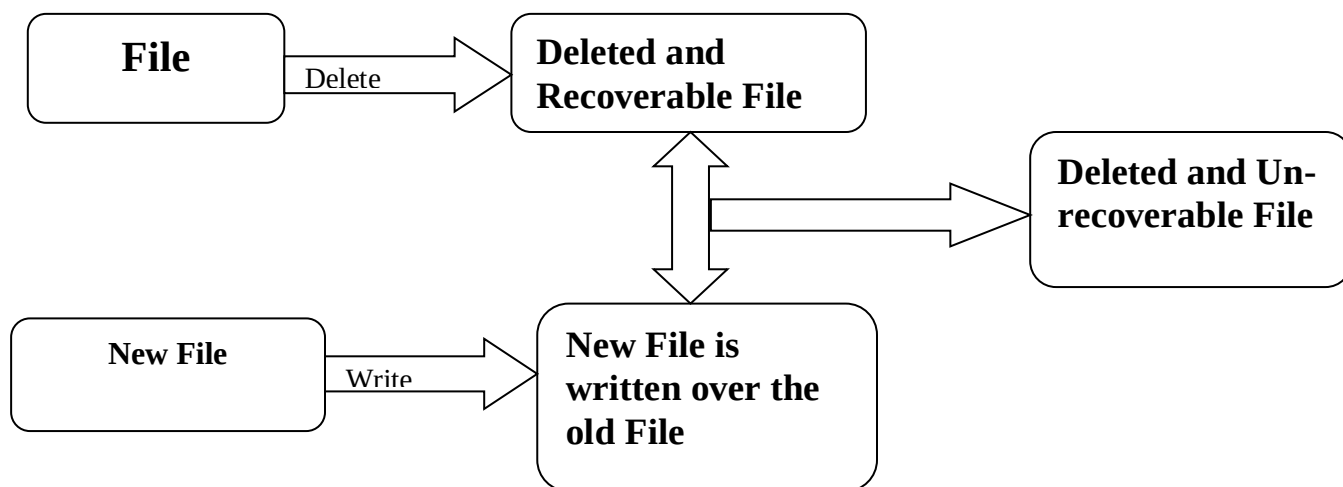
Data recovery techniques

instant recovery, also known as recovery in place, tries to eliminate the recovery window by redirecting user workloads to the backup server.

A snapshot is created so the backup remains in a pristine state and all user write operations are redirected to that snapshot; users then work off the backup virtual machine (VM) and the recovery process begins in the background. Users have no idea the recovery is taking place, and once the recovery is complete, the user workload is redirected back to the original VM.

One way to avoid the time-consuming and costly process of data recovery is to prevent the data loss from ever taking place. Data loss prevention (DLP) products help companies identify and stop data leaks, and come in two versions: stand-alone and integrated.

- Stand-alone DLP products can reside on specialized appliances or be sold as software.
- Integrated DLP products are usually found on perimeter security gateways and are useful for detecting sensitive data at rest and in motion.



Exercise:

1. Use Autopsy to analyze the Android image of your phone and explore the recovery options.
2. Recover deleted SMS messages on your Android device using appropriate techniques.
3. Recover contacts using Google account features.

Questions:

1. How did you use Autopsy to analyze the Android image, and what options does it provide for file recovery?
2. What specific techniques did you employ to recover deleted SMS messages on your Android device?
3. Describe the steps you followed to recover contacts using your Google account.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 6

Multimedia Forensics

Objective:

Develop skills in multimedia forensics by utilizing tools such as Autopsy and Exif Reader to extract metadata from images.

Theory:

Multimedia forensics involves the analysis of digital media, including images, to extract valuable information, such as metadata, that can be crucial for investigations.

Multimedia forensics is a branch of forensics that works hand in hand with computer forensics. While Computer forensics can be defined as the branch of forensics that deals with the application of investigative techniques to extract and preserve evidences from a digital device submitted to the court of law. Multimedia forensics on the other hand, deals with analyzing the extracted evidence. The analysis includes scientific evaluation of the digital evidence, to maintain the integrity of evidence, to find the source of the digital evidence and to authenticate the same

The tools used by the multimedia forensic scientist

There are many tools that are in use today by the multimedia forensic experts to examine the digital evidences.

For audio recordings that are submitted as the evidence, tools like FTK Imager, Gold wave, Multispeech, Computerized speech Lab, Acustek, Audacity etc. are commonly used.

For the examination of other digital evidences like images and videos, tools like Amped five, Amped Authenticate, Kinesense, MIDAS, Adobe Photoshop, Brief cam, Pinnacle, clear Id etc. are commonly used. In the case of video footages provided for the examination, the video clip is converted into frames and then analyzed

Multimedia Forensics Categories

Web content isn't just constrained to textual format; it arrives in a variety of assortments, so the criminological methodologies created to examine them should likewise fluctuate in scope. The objective here is to analyze the pictures, text, audio as well as video, so as to create a bit of coherent Forensic evidence.

1. Active Image Authentication and
2. Passive Image Authentication.

Active as well as passive image verification are additionally branched into different categories.

Active Image Authentication

In this strategy, a known validation code is inserted in the image at the time of image formation or sent with the image for accessing its genuineness at the receiver end. Examination of this code confirms the originality of the image.

The active image authentication methodology is sub-divided into two categories:

1 Digital Watermarking

In multimedia forensics, the watermarking is a method utilized to recognize the authentic source of a particular image or picture along with its user verification.

This watermark is a specific code of different characters which is entrenched in a digital text file or document file, audios, videos, images, or computer software program that offers the fine points to the investigating officer to observe more in relation to the images they came upon. This process is used to make it more easier for copyright holders to identify the individuals who tried to violate these rights.

Moreover, a forensic watermark is able to alert an original user in the consequence that they unintentionally receiving an illegal document or software program.

2. Digital Signatures

Digital signatures are another category of active authentication methodology.

These signatures are similar to the handwritten signatures but they come out in an electronic form.

The major role of these signatures is to encode the contents of a multimedia file or document, and consequently allowing those specific software applications to discover if there is any kind of alternation or tampering done in either ways.

From a legal viewpoint, this furthermore allows the maker to show that a document existed at a certain date and time.

Downsides of Active image authentication:

- The authentication code needs to be inserted in the image at the time of recording by utilizing extraordinary tools therefore earlier information concerning the picture or image becomes very important.
- This method necessitate a digital watermark or a computerized signature to be made accurately while recording the image, which limits its ability to handle specially equipped digital devices which constrains its capacity to deal with uniquely prepared computerized gadgets.
- As most of the pictures on the Internet, today don't have a watermark or a digital signature, which has made this process to mull over some supplementary techniques.

Passive Image Authentication

This method is also recognized as image forensics which requires only image itself for accessing its veracity without any secondary information for example a signature or watermark from the sender.

This type of authentication based on the hypothesis that even if alteration in the image may not depart any visible trace but still they are expected to modify the primary statistics which shows that forgeries with the digital evidence may agitate the original properties as well as quality of the picture, even if there is no any visual clue has been found.

Passive image authentication is also subdivided in 2 categories

1. Forgery-Type Dependent

This method is used to identify only some specific type of forgeries such as copy-paste and splicing of the image which are relented on the nature and type of forgery carried out on the

image. It is again sub-classified into two categories:

2. Copy-Move Detection

This is a well-known photograph altering method due to the effortlessness with which it tends to be done.

It includes replicating a few locales in the picture and moving it to some other region in the picture. Since the replicated area has a place on the same picture so the dynamic range and shading are well matched with the picture.

While detecting this kind of forgeries in the image some post-processing technique such as obscuring is utilized to diminish the impact of fringe abnormalities between the two pictures.

3. Image-Splicing Detection

This includes the assimilation of 2 or more photographs that significantly modify the original image to create a falsified image. Please note that when combining images with different backgrounds, it becomes difficult to make the borders indistinguishable. Detecting spliced images is a complex task involving the following methods:

- Composite regions are studied by various methods.
- The presence of abrupt changes between different areas that combine to create a composite image and their backgrounds provide valuable traces for detecting splicing in the image in question.

Exercise:

Use Autopsy and Exif Reader to extract metadata from JPG, PNG, and BMP images on your computer.

Questions:

1. How did you use Autopsy to extract metadata from images?
2. What information is typically found in the metadata of JPG, PNG, and BMP images?
3. Why is extracting metadata important in the context of multimedia forensics?

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 7

Cloud Forensics

Objective:

Develop skills in cloud forensics using the WinHex editor to perform operations related to Google Drive, such as access time, program run count, and detailed user transactions.

Theory:

Cloud forensics involves the examination and analysis of digital evidence stored in cloud services. Tools like WinHex can be employed to perform operations on cloud-related data.

Cloud forensics refers to the use of forensic techniques to investigate cloud environments. When unlawful or criminal behavior has occurred using the cloud as a medium, cloud forensics experts use their skills and knowledge to detect the individuals or groups responsible. Cloud forensics encompasses users of the cloud, both victims and perpetrators. For example, a company using cloud servers might be the victim of a data breach or denial of the service incident. Criminals themselves might also use the cloud to launch an attack.

The investigation process in cloud forensics can be divided into four main stages:

- **Identification**
- **preservation**
- **analysis**
- **presentation.**

Identification

The first stage of the investigation process involves identifying the cloud-based systems and applications that are relevant to the investigation. This may include identifying the cloud service provider, the specific services and applications used, and the data that is stored or transmitted through these systems.

Preservation

The preservation stage involves taking steps to preserve the digital evidence, so that it can be analyzed and presented in a court of law, if necessary. This may include capturing and storing data from the cloud-based systems and applications, as well as documenting the steps taken during the investigation process.

Analysis

The analysis stage involves using forensic tools and techniques to analyze the digital evidence and identify any relevant information. This may include examining log files, network traffic, and metadata, as well as using data recovery tools to recover deleted or damaged data.

Presentation

The final stage of the investigation process involves presenting the findings in a clear and concise manner, so that they can be used as evidence in a court of law, if necessary. This may include creating reports, presenting visual aids, and testifying in court as an expert witness.

Exercise:

1. Use WinHex to find out the last time you accessed Google Drive on your computer.
2. Determine the number of times the Google Drive program has been run on your computer.
3. Extract a detailed list of a user's cloud transactions, listing the create, modify, and delete dates and times.

Questions:

1. How did you use WinHex to find the last access time of Google Drive on your computer?
2. What steps did you take to determine the number of times the Google Drive program has been run?
3. Describe the process of extracting a detailed list of user cloud transactions and noting create, modify, and delete dates and times.

Assignment Evaluation**0: Not Done []****1: Incomplete []****2: Late Complete []****3: Need Improvement []****4: Complete []****5: Well Done []****Signature of Instructor**

Assignment 8

Virtual Machine and Network Forensics

Objective:

Gain practical experience in virtual machine and network forensics using VMware Workstation Player, FTK Imager Lite, tcpdump, and Wireshark.

Theory:

Virtual machine and network forensics involve the examination of evidence within virtualized environments and the analysis of network activities for investigative purposes.

Virtual machines are not new and have been in use for well over a half century. The fundamental concept of a virtual machine revolves around a software application that behaves as if it were its own computer. The VM application (“guest”) runs its own self-contained operating system within the actual machine (“host”). This virtual operating system can be of almost any variant of design. Perhaps put more simply, it can be described as a virtual computer running inside a physical computer.

One of the benefits of virtual machines is the ability for a virtual machine to operate on nearly any underlying hardware and software configuration. In this manner, there is an ease of flexibility of sharing and duplication of virtual machines for many purposes, such as software testing. Additionally, one host machine (the actual computer) can run multiple guest machines (virtual machines) at the same time.

The following is a listing of the files associated with a VMware virtual machine. With other VM applications, the files may be fewer or even more, so it is imperative to be aware of the associated file types when dealing with different types of virtual machines other than VMware. The existence of only one of these files can indicate that a virtual machine may have existed on the media being examined.

1. .Log files -Simply a log of activity for a virtual machine.
2. .VMDK- This is the actual virtual hard drive for the virtual guest operation system, which may be either a dynamic or fixed virtual disk. With dynamic disks, the disk will start small and grow to a predetermined limit. A fixed disk does not change size.
3. .VMEM -A backup of the virtual machine’s paging file which only exists if the VM is running or has crashed.
4. .VMSN – These are VMware snapshot files, named by the name of the snapshot. A VMSN file stores the state of the virtual machine when the snapshot is created.
5. .VMSD-A VMSD file contains the metadata about the snapshot.
6. .NVRAM- This is the file that stores the BIOS information for the virtual machine.

7. .VMX- This is the configuration file for a virtual machine, such as the operating system, disk information, etc This is a simple text file that can be easily edited.
8. .VMSS-This is the 'suspended state' file, storing the state of a suspended virtual machine.
9. .VMTM-This is configuration file containing team data.
10. .VMXF-If a virtual machine is removed from a team, this configuration file remains.

Exercise:

1. Use VMware Workstation Player and FTK Imager Lite to examine your own system for evidence of a virtual machine (VM).
2. Acquire a forensic image of the host computer (physical machine) and collect network logs.
3. Extract and examine files associated with VMs, including log files, to determine the timeline of a crime or incident and find relevant information such as accessed websites and network files.
4. Utilize the tcpdump command-line program to examine network traffic.
5. Use Wireshark to generate a list of the top 10 websites users in the network are visiting.

Questions:

1. How did you use VMware Workstation Player and FTK Imager Lite to identify evidence of a virtual machine on your system?
2. Describe the steps involved in acquiring a forensic image of the host computer and collecting network logs.
3. What information can be obtained by examining files associated with VMs, especially log files, in the context of investigating a crime or incident?
4. How did you use the tcpdump command-line program to examine network traffic?
5. What insights were gained by using Wireshark to generate a list of the top 10 websites visited by users in the network?

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

